Statistics and Experimental Design

#1 - Introduction

By:
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Introduction

- Lecturer Profile
- Course Profile

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Brief Description of the Course

- The aim of this course is to help students design data collection through experiments and be able to analyze and interpret experimental data.
- Theoretical material is delivered through lectures and discussions.
- Likewise, data analysis material is delivered through discussions and discussions as well as practicums in class with and without software.
- In addition, students are given assignments, both independently and in groups, to identify the application of various designs to real problems, either through independent observation or through field studies in government agencies or industries where there are research and development activities.
- The aim of this assignment is to train students to be able to manage and work in teams and be responsible for the results of independent and group work.

Lecture Regulations

1. Minimum attendance: 80% (as a mandatory requirement to pass).
2. Students not only attend but also have to be actively involved in lectures.
3. Assessment components:
   a. Participatory activities : 55%
   b. Tasks : 10%
   c. Quiz : 5%
   d. Mid exam : 15%
   e. Final exam : 15%
4. Students who are proven to have committed **academic violations** will receive a **maximum grade of D**.

Topics

Week 1: Introduction to statistics

Week 2: Descriptive statistics

Week 3: Probability concept

Week 4: Data collection/sampling techniques

Week 5: Hypothesis testing

Week 6: ANOVA test

Week 7: Correlation and Regression

Week 8: Mid Exam

Week 9: Concept of experimental design

Week 10: Complete Random Design

Week 11: Completely Randomized Block Design

Week 12: Latin square

Week 13: Common factorial (2k and 3k)

Week 14: Response surface

Week 15: Taguchi

Week 16: Final Exam

Learning Method

- Combination of lectures and case study
- Hybrid learning (Online and in-Class)
- Working in Group
- In-depth study of lecture and practical material is carried out by giving assignments to students.

References

- Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. (1993). Probability and statistics for engineers and scientists (Vol. 5). New York: Macmillan.
- Montgomery, D.C, 2005. Design and Analysis of Experimentals. 8th edition. New York: John Wiley dan Sons Inc.

Introduction to Statistics

Why Study Statistics???

Statistic and Statistics

- To start with the introduction on Statistics, we should know what it means. These words, Statistics and statistic are used very commonly in our day to day conversation but we don’t know the difference between them. Why should we start with the difference between Statistic and Statistics? Because, the difference uncovers the definition of statistics and clears us from the confusion.
- There is a very common misinterpretation about the word in singular sense- “STATISTIC” and in plural sense “STATISTICS”.
- To deviate a bit more, and to know the difference, we first need to know what a SAMPLE is and what is a POPULATION.

What is the difference ???

Difference between sample and population

- When we think of Population, the first thing that comes to our mind is population of a country, a state or any particular geographical area. In statistics, Population is just that.
- Population consists of ALL the people or objects or items that are included in our study or the topic of our research.
- E.g. I want to study the effectiveness of extra classes in school students of Jakarta.
- My population consists of ALL SCHOOL GOING CHILDREN IN JAKARTA.

Difference between sample and population

- It is almost impossible and impractical for the researcher to reach the whole population. No doubt that the best results are obtained if each and every item of our population is covered but to be feasible and quick, researchers RANDOMLY select a PART of the population which is called a SAMPLE.
- When we obtain the results from the Population, it is called a Parameter and when we obtain results from a Sample, it is called Statistic.

Why we need sample???

Limited Resources

Time Issue

Observation sometimes destructive

Impossible to observe the whole

GIGO

Garbage in Garbage Out

Coming back to the difference

Statistics

The process of designing, comparing, interpreting and analyzing **data.**

Statistic

The score of each individual or a singular data

What happened with Data?

- Data → A form of repeated recording of the characteristics of an object.
- Each individual is recorded with a certain **identity.**
- Each characteristic is recorded into a certain **variable.**

Table 1. A dataset with variables X, Y and Z.

n	X	Y	Z
1	x_1	y_1	z_1
2	x_2	y_2	z_2
3	x_3	y_3	z_3
4	x_4	y_4	z_4
5	x_5	y_5	z_5
6	x_6	y_6	z_6
7	x_7	y_7	z_7
8	x_8	y_8	z_8
9	x_9	y_9	z_9
10	x_{10}	y_{10}	z_{10}

- A table with 10 rows and 3 columns.
- A dataset of 10 identities, and 3 variables.
- Variable names are generally denoted by capital letters, while variable values are denoted by lowercase letters accompanied by a subscript that states the identity.

Statistics

Statistics → Methods

- A method of processing data obtained from repeated operations.

The operation is obtained through:

- Observation
- Measurement
- Trial/Experiment

The Use of Statistics

- Descriptive Statistics
- Statistical Inference

Descriptive Statistics

Suppose a metal factory wants to analyze the quality of their aluminum bar production during the last month.

They used **descriptive statistics** to describe the production characteristics, such as the average weight of aluminum ingots, variations in thickness, and distribution of ingot lengths.

This analysis helps them understand in detail how the production process works and identify areas where improvements may be needed.

Statistical Inference

On the other hand, the factory may want to draw broader conclusions about the quality of their production based on the samples taken.

For example, they may want to know whether a statistically significant difference in tensile strength between two different types of aluminum bar can be accounted for by differences in their quality.

To do this, they will use **Statistical Inference**, such as hypothesis testing, to determine whether the differences they see are not simply the result of natural variability in production, but may represent significant differences in quality.

Descriptive vs Inference

Descriptive statistics are used to **summarize** the observed data

The focus in on describes the characteristics of a data set. It is a simple technique to describe, show and summarize data in a meaningful way.

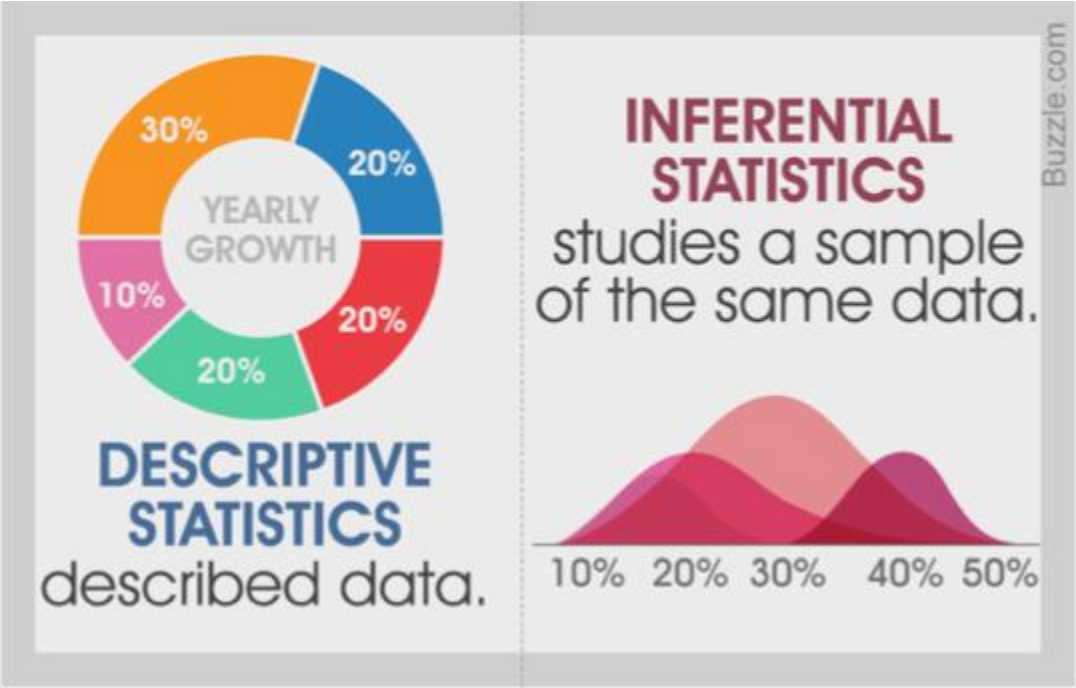
To obtain insights and visualize the data rather than simply pouring through sets of raw numbers.

Statistical inference is used to **make conclusions** or inferences about a larger population based on an observed sample.

The focus is on making predictions about a large group of data based on a representative sample of the population.

The accuracy of inferential statistics depends largely on the accuracy of sample data and how it represents the larger population.

Descriptive vs Inference



Descriptive vs Inference

Descriptive

- Gives information about raw data regarding its description or features.
- To describe a situation.
- Helps to organize, analyze and present data in a meaningful manner.
- Use charts, graphs, and tables.

Inference

- Draw inferences about the population by using data extracted from the population.
- To explain the probability of occurrence of an event.
- Helps to compare data, make hypotheses and predictions.
- Use probability methods.

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